

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

WORK EXPERIENCE

NATIONAL UNIVERSITY OF SINGAPORE *Postdoc under Kostya Novoselov* 2022–present

Wyckoff Transformer for generating symmetric crystals [ICML 2025](#) · [code](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference throughput by 4 orders of magnitude, dropping generation costs to 0.05 GPU ms per structure.
- Implemented the model in PyTorch; production-ready package published on [PyPI](https://pypi.org/project/wyckoff-transformer/)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

ML for predicting properties of defects in 2D crystals [paper 1](#) · [paper 2](#) · [patent](#) · [code](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference workflows, reducing memory usage by 4x and GPU operations by 8x
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

CONSTRUCTOR GROUP *Machine Learning Consultant* 2021–present

CERN [YANDEX → HSE UNIVERSITY] *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models for fast simulation [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately 10^5 speed-up compared to the ab-initio simulator.

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

EDUCATION

Higher School of Economics (HSE University) 2016–2020
PhD in Computer Science, supervisor [Andrey Ustyuzhanin](#); [Machine Learning for particle identification in the LHCb detector](#).

Sapienza – Università di Roma 2016–2020
PhD in Physics (double degree with HSE), supervisor [Barbara Sciascia](#).

Product Management course at Yandex 2018
Focused coursework in product strategy and execution.

Yandex School of Data Analysis 2013–2015
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.
BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

TEACHING & OUTREACH

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

OTHER

- Aspiring stand-up comedian; bombed more than 20 times at open mics.
- International Junior Science Olympiad 2008 in Korea, silver medal.

SKILLS

- **ML & AI** – Structured, non-text, domain-specific data
- **Python** – Research coding
- **PyTorch** – Deep learning
- **C++** – Production
- **Linux** – Administration
- **Research Writing & Speaking**
- **Physics** – Deep domain understanding
- **Leadership & Mentorship**

SELECTED PUBLICATIONS

1. **Kazeev, Nikita**, and Ian Babich. [Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments](#). Forecasting as a New Frontier of Intelligence Workshop at ICML 2026
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Bundenyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials](#). AI for Science Workshop at ICML 2026
3. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. [Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement](#). arXiv preprint arXiv:2605.14759, May 2026.
4. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. [LeMat-GenBench: Bridging the gap between crystal generation and materials discovery](#). AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
5. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
6. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). npj Comput Mater 9, 113 (2023).
7. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. Alphabetic order, I'm the corresponding author.